

BANKING CUSTOMER ANALYTICS & DASHBOARD — PROJECT REPORT

An end-to-end customer data analysis project using Python, PostgreSQL/SQL and Power BI

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1. Project Overview

This project analyzes a retail banking customer dataset covering demographic, account and financial-relationship information for 3,000 clients. The work spans the full analytics pipeline: cleaning and exploring the raw data in Python, storing it in a PostgreSQL database for SQL-based access, and presenting the results through an interactive three-page Power BI dashboard covering customer loans and deposits.

The goal is to turn raw customer records into a clear picture of who the bank's clients are, how their loan and deposit balances vary by branch, occupation, nationality and income level, and to package these insights into a dashboard that business stakeholders can filter and explore on their own.

2. Objectives

- Clean and standardize the raw customer banking dataset and verify data quality (completeness, types, duplicates).
- Engineer an income-band feature to segment clients into Low, Medium and High income groups.
- Explore demographic, relationship and financial variables to uncover patterns and relationships in the client base.
- Persist the cleaned dataset in a relational (PostgreSQL) database to enable SQL-based querying.
- Design an interactive Power BI dashboard that lets stakeholders filter and analyze loans and deposits by branch, gender, occupation, nationality and income band.

3. Tools & Technologies

Category	Tools Used
Data Handling & EDA	Python — pandas, NumPy
Data Visualization (EDA)	Matplotlib, Seaborn
Development Environment	Jupyter Notebook
Database	PostgreSQL
Database Connectivity	SQLAlchemy, psycopg2
Query Language	SQL
Dashboard & Reporting	Power BI Desktop

4. Dataset Overview

Metric	Value
Source file	Banking.csv

Metric	Value
Total records	3,000 clients
Total attributes	25 columns
Missing values	0 — fully complete dataset
Database / table	banking_data.customer_banking_data (PostgreSQL)
Engineered feature	income_band — Low / Medium / High, derived from Estimated Income

Data Dictionary (grouped by category)

Category	Fields
Client profile	Client ID, Name, Age, Location ID, Joined Bank, Banking Contact, Nationality, Occupation, Gender ID
Relationship & risk	Fee Structure, Loyalty Classification, Risk Weighting, Branch ID (BRId), Investment Advisor ID (IAId)
Income & savings	Estimated Income, Income Band, Superannuation Savings
Credit & loans	Amount of Credit Cards, Credit Card Balance, Bank Loans, Business Lending
Deposits & assets	Bank Deposits, Checking Accounts, Saving Accounts, Foreign Currency Account, Properties Owned

5. Methodology

- **Data ingestion** — Loaded the raw Banking.csv file into a pandas DataFrame and reviewed its structure with .head(), .info() and .describe().
- **Cleaning & validation** — Standardized all column names to lowercase, underscore-separated form for consistency; confirmed zero missing values across all 3,000 records with .isnull().sum().
- **Feature engineering** — Created the income_band column by splitting Estimated Income into three equal-sized groups (Low / Medium / High) using quantile-based binning (pd.qcut).
- **Data warehousing** — Wrote the cleaned DataFrame into a PostgreSQL table (customer_banking_data, database banking_data) via SQLAlchemy, making the dataset queryable with SQL.
- **SQL verification** — Used SQL (see banking_db.sql) to query and verify the loaded table, e.g. SELECT * FROM customer_banking_data LIMIT 20.
- **Exploratory analysis** — Examined categorical variables with count plots (overall, and split by gender and nationality), numerical variables with histograms, and inter-variable relationships with a correlation heat map.
- **Dashboard design** — Built a 3-page Power BI report (Home, Loan Analysis, Deposit Analysis) with slicers, KPI cards and charts for interactive exploration.

6. Exploratory Data Analysis — Key Insights

6.1 Client Demographics

- Client ages range from 17 to 85 years, averaging around 51 — a broad, evenly-spread customer base rather than one concentrated age group.
- Gender is almost evenly split across the client base (1,512 vs. 1,488 clients in the two recorded categories).
- Nationality is dominated by European clients (1,309 — 43.6%), followed by Asian (754 — 25.1%), American (507 — 16.9%), Australian (254 — 8.5%) and African (176 — 5.9%).

6.2 Relationship, Loyalty & Risk

- Fee Structure skews toward the High tier (1,476 clients — 49.2%), with Mid (962 — 32.1%) and Low (562 — 18.7%) making up the rest.
- Jade is the largest Loyalty Classification (1,331 — 44.4%), ahead of Silver (767), Gold (585) and Platinum (317) — a natural upsell funnel toward higher tiers.
- 64% of clients hold exactly one credit card, 25.5% hold two, and 10.4% hold three.
- Risk Weighting (1–5 scale) is concentrated at the lower end: level 2 is most common (1,222 clients — 40.7%) and level 1 next (836 — 27.9%); only 16% of clients sit at the higher-risk levels 4–5.
- Clients are fairly evenly spread across 0–3 properties owned (roughly 700–780 clients per group).
- Branch 3 (BRId) holds the largest share of clients (1,352 — 45%), well ahead of branches 1, 2 and 4 (493–660 each).

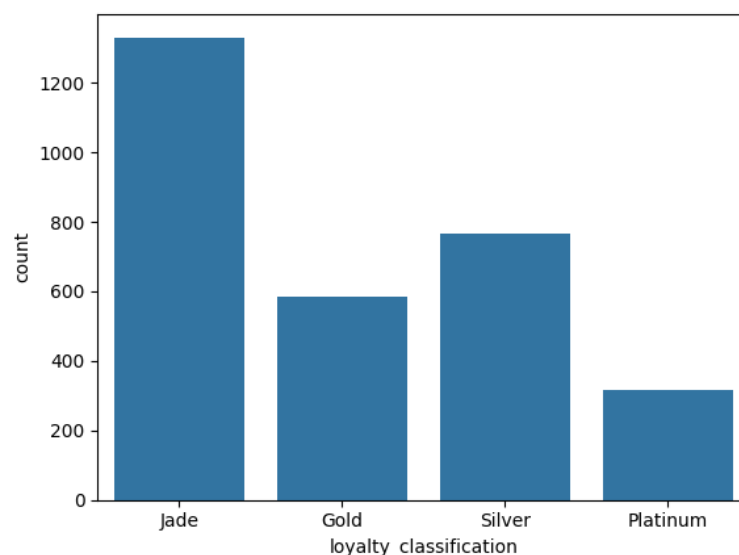


Figure 1 — Client distribution by Loyalty Classification

6.3 Financial Variables & Correlations

- All nine financial variables — income, savings, credit card balance, loans, deposits, checking, saving, foreign-currency and business-lending balances — are right-skewed: most clients cluster at lower values with a long tail of high-value clients.

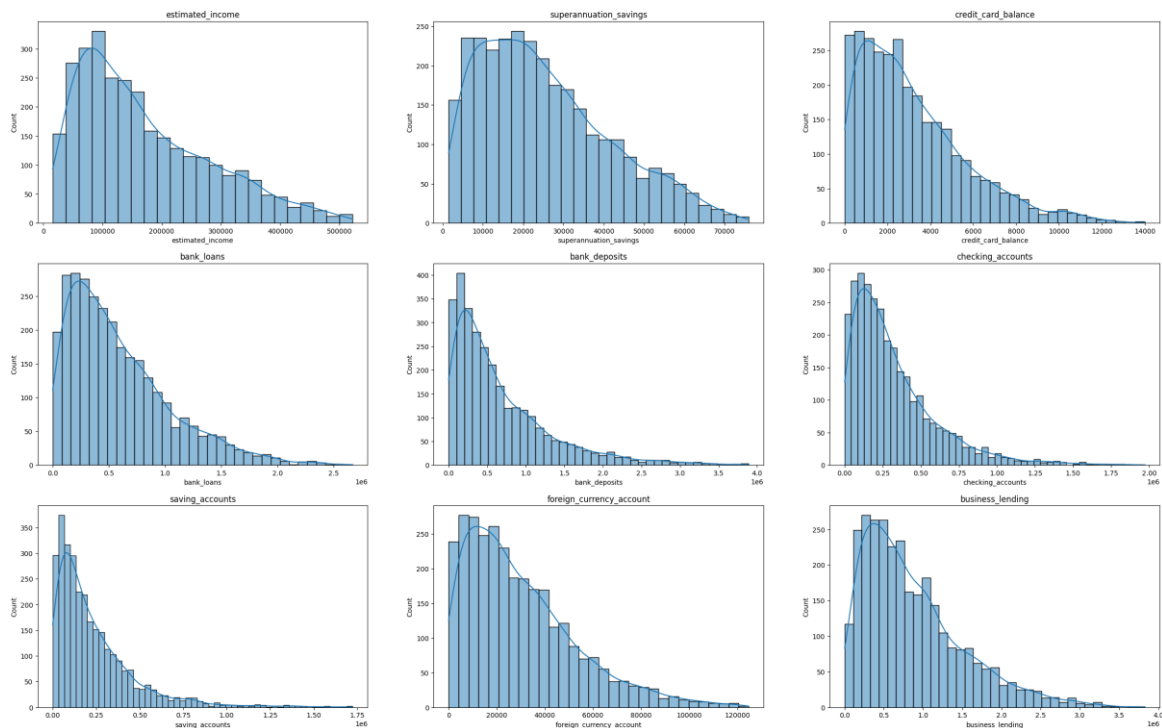


Figure 2 — Distribution of numerical financial variables

- Bank Deposits correlates strongly with Checking Accounts (0.84) and Saving Accounts (0.75), which makes sense since both feed into a client's total deposit balance.
- Every other pair of financial variables shows only mild-to-moderate correlation (0.17–0.44) — each metric captures a fairly distinct part of a client's financial profile, which is useful for segmentation rather than redundant.



Figure 3 — Correlation matrix of financial variables

9. Conclusion

This project demonstrates a complete, practical analytics workflow: cleaning and exploring raw banking data in Python, storing it in a relational database for SQL access, and delivering the results through an interactive Power BI dashboard. Together, the notebook, database and dashboard give a consistent, end-to-end view of the client base — from individual client attributes to branch-level loan and deposit trends — that supports customer segmentation, relationship management and risk-monitoring decisions.

10. Future Scope

- Build a predictive model (e.g., logistic regression or a tree-based classifier) to flag clients at higher risk of default or churn.
- Automate the CSV-to-PostgreSQL load with a scheduled pipeline so the dashboard refreshes on a regular cadence.
- Add drill-through pages from each chart to a client-level detail view.
- Apply row-level security in Power BI so branch managers see only their own branch's data.
- Extend the analysis with clustering (e.g., k-means) on financial variables for a data-driven customer segmentation model, beyond the rule-based income band.